THE ATMOSPHERE 1

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The atmosphere is the name for the air (gases) surrounding the Earth, reaching from the Earth's surface to the edge of space. Scientists have divided the atmosphere into various layers (usually based on temperature).

Figure 1 (not to scale) shows a simplified view of the various atmospheric layers. There are no distinct boundaries and overlapping between different layers does occur; this is why textbook values can vary slightly.

Some areas exhibit specific physical or chemical characteristics, but don't quite fit in the diagram shown. For example, the ozonosphere is a region where ozone is produced and the ionosphere contains charged particles (ions) that allow radio signals to be bounced from one country to another.

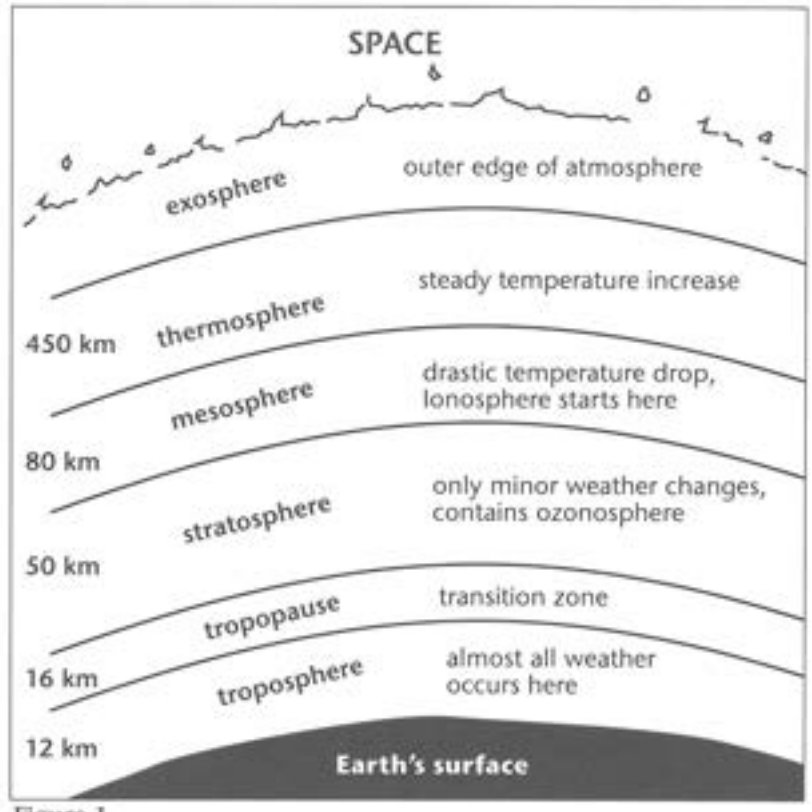


Figure 1

1 Draw up a table to summarise the information above and in figure 1 about the atmospheric layers. Use the headings shown below.

Name of layer	Distance from surface (km)	Features of layer

Air pressure A huge amount of air is pressing down on us at the Earth's surface. Although we cannot see air pressure, it can be measured. Air pressure varies with the height above ground and the weather, but a standard pressure at sea level is 1 atmosphere (equal to 101.3 kilopascals, 760 mm of mercury, 14.7 pounds per square inch and 1013 millibars).

2 The weather forecast includes air pressure readings along with temperatures and other features of the day's weather. Record your local air pressure readings, temperature and weather conditions for five consecutive days. Draw conclusions as to how, or if, connections can be made between air pressure, temperature and weather conditions.

Composition of the atmosphere The three main gases present in the atmosphere are nitrogen (N_2, 78.08%), oxygen (20.95%) and argon (Ar, 0.94%). Water vapour (H_2O) is also present in variable amounts depending on weather conditions. The remaining gases, although present in minute amounts, still form an important part of the atmosphere. These include carbon dioxide (CO_2), neon (Ne), helium (He), methane (CH_4) and krypton (Kr). Other gases present in variable amounts include ozone (O_3) and sulfur dioxide (SO_2).

3 Draw up a table to summarise the information about atmospheric gases. Use the headings shown below.

Name of gas	Formula	Element or compound	Amount in air (%)

4 If you were to ascend through the various atmospheric layers, would you be able to see distinct boundaries? Explain your answer.

Research 5 Methane and carbon dioxide are known as *Greenhouse gases*. Find out what is meant by this term and why the word *greenhouse* is used.

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Separation of gases in the atmosphere Air is a source for many gases used in the chemical industry. Gases can change to liquids through a process of compressing and cooling. The difference in boiling points of gases can be used to separate out the different gases in air. In this process, air is liquefied and then gradually allowed to warm up. Each component changes back to a gas when it reaches its boiling point. (Table 1 shows the boiling points of some gases.) Each gas is extracted as it evaporates out of the liquid at its specific boiling point. This process of **fractional distillation** does not give pure gases initially; further stages are required.

6 Two common ways of describing temperature are the Celsius (°C) and Kelvin (K) scales. Explain how to convert degrees Celsius into Kelvins.

7 Consider a liquid sample of air that been has cooled to –200°C. If the only gases present are nitrogen, oxygen and argon, in what order would they turn to gas as the mixture warmed?

Gas	Boiling point (°C)	Boiling point (K)
Nitrogen	–196	77
Oxygen	–183	90
Argon	–186	87

Table 1

Nitrogen and its compounds Nitrogen has no taste, no smell, no colour and is relatively unreactive. It exists as diatomic molecules.

8 Explain what is meant by a *diatomic molecule*.

Nitrogen was first isolated around 1772 by several scientists independently. These were Daniel Rutherford (Scotland), Joseph Priestley (England), Henry Cavendish (England) and Carl Wilhelm Scheele (Sweden). Antoine Lavoisier (France) was the first person to realise that nitrogen was an element.

Research **9** Find out one more fact about each of the scientists mentioned above.

Nitrogen has many uses that rely on the fact that it doesn't support oxidation or combustion. Nitrogen is used to replace air in the packaging of some foods (to prevent oxidation). In its liquid form it is used to 'flash freeze' foods. In this process liquid nitrogen is sprayed onto foods and they freeze almost immediately. The nitrogen then evaporates. The frozen food, on thawing, retains most of the properties of fresh food because the freezing process is so quick. Liquid nitrogen is also used to preserve biological samples.

10 A manufacturer of premium potato chips filled each packet with nitrogen gas (as well as chips!) before sealing the packet. Why?

11 Give two uses of liquid nitrogen.

Nitrogen compounds are a major factor in maintaining life. Nitrogen is an important component of amino acids (which form proteins) as well as the giant molecules DNA and RNA, the genetic material that carries the 'blueprint' information for living things. Nitrogen has to be returned to the soil after crops have been harvested. A major use of nitrogen is in the Haber process. This process produces ammonia (NH_3) from hydrogen and nitrogen. Ammonia is used to make nitric acid; this in turn is the raw material in producing nitrates for use in dyes, explosives and fertilisers. Certain species of **bacteria** live either separately or associated with certain plants (legumes) and are known for **nitrogen fixation**.

12 Write a balanced formula equation for the formation of ammonia from nitrogen and hydrogen.

Research **13** Use reference materials to give a brief definition for *nitrogen fixation*.

14 Name one legume.

Oxygen This gas was discovered by Joseph Priestley in 1774. He heated mercury oxide (MgO), forming mercury and oxygen. Oxygen was further investigated by Antoine Lavoisier who discovered many of its properties and gave oxygen its name.

15 Write a balanced formula equation for the thermal decomposition of mercury oxide.

Oxygen is similar to nitrogen in that it has no colour, taste or odour. It is a very reactive element and is needed by animals for respiration. The greatest industrial use of oxygen is in blast furnaces producing steel. Other uses include life support for hospital patients and oxy-acetylene welding. Acetylene (ethyne) has the formula C_2H_2 and when burned with oxygen produces carbon dioxide and water. Until 1961, oxygen was used as the standard for atomic masses. The isotope oxygen–16 was assigned a mass of 16.000 amu.

16 Write a balanced formula equation for the burning of acetylene in oxygen.

Research **17** Which isotope do scientists now use as the standard for atomic masses?

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